

Design Note — Lexical & Semantic Retrieval on EB-NeRD and MIND

CS4.406 Information Retrieval & Extraction, Assignment 1. Code: `aayush18602/IRE_Assignment_1`.

1. What We Built

A dataset-agnostic pipeline over EB-NeRD (Danish) and MIND (English), sharing one schema and one code path for both:

- **Q1 — Pipeline:** `clean_ebnerd/clean_mind` parse both datasets into one schema (articles, impressions, history; ids cast to string so 9778623 and "N55689" compare equal). `temporal_split` cuts train/val/test at quantiles of impression time (never random), with a leakage assertion built into the function itself. Feature store: `article text/category/entities`, user click history + `history_asof()/recent_history_asof()` for leak-safe, as-of-impression-time history truncation — reused by every later Q.
- **Q2 — BM25:** hand-rolled inverted index + Okapi scoring (`src/ire_a1/bm25.py`), not a library wrapper, per the assignment's explicit ask. Query = titles of the user's N=10 most recent clicks, as-of the impression's own timestamp.
- **Q3 — Embeddings:** mean-pooled transformer embeddings over title+abstract, ANN via FAISS flat (exact) index. Two variants run deliberately (§3): `raw_xlm-roberta-base`, and `paraphrase-multilingual-mpnet-base-v2` (fine-tuned for similarity, same XLM-R backbone).
- **Q4 — Eval harness:** AUC/MRR/nDCG@5/nDCG@10 (sklearn, official-style — re-ranking each impression's own shown candidates, not a full-catalog list), diversity/novelty/coverage, cold/warm slicing, bootstrap 95% CIs on every metric.
- **Q5 — Codabench submission:** both BM25 and embeddings re-ranking, run at the actual large-test-set scale (13.5M EB-NeRD / 2.37M MIND impressions).
- **Q9 — Anti-gaming:** a deliberately-constructed leaky-feature ablation (§4) and a dedicated no-leakage test suite (`tests/test_no_leakage.py`).
- **Testing:** 51 unit/integration tests (pytest), covering the split's leakage invariant, BM25/ANN scoring correctness, eval-metric edge cases, submission-format correctness, and the Q9 leaky-vs-safe popularity scoping.

2. Design Choices & Alternatives Considered

Compute split (local CPU vs. Kaggle GPU). This machine has no working CUDA driver. Rather than default everything to Kaggle, we checked what actually needs a GPU: BM25 is CPU-only by nature; loading provided embeddings needs no GPU either. Only *computing our own* transformer embeddings does. So: Kaggle only for embedding computation, everything else — including the 13.5M-row Q5 submission scoring — local.

BM25 index structure. Postings stored as `term -> {doc_idx: tf}` (not `term -> [(doc_idx, tf), ...]`). This one choice enabled two different access patterns cheaply: `query()` scans all postings for full-catalog top-K (Q2/Q3 candidate generation), while `score_candidates()` does the reverse — loop over the ~15–30 *given* candidates first, touching zero documents outside that set. This was not a minor optimization; it's what made Q5 tractable at all (§4).

Raw vs. fine-tuned embeddings. The assignment names "BERT/XLM-RoBERTa" as the embedding choice. We first ran plain `xlm-roberta-base`, mean-pooled — a vanilla MLM checkpoint never trained for similarity. When it lost to BM25 on both datasets, we added a second variant, `paraphrase-multilingual-mpnet-base-v2` — architecturally still an XLM-R-base checkpoint, just fine-tuned via multilingual distillation — as a fairer semantic baseline, not a replacement.

Percentile, not fixed, cold/warm threshold. A fixed `history_length < 5` cutoff gave an *empty* cold slice on EB-NeRD — its small/demo tiers are pre-filtered by the dataset's own creators to a minimum history length of exactly 5 for every user (verified via `.describe()` before trusting a fixed threshold). Switched to a 25th-percentile-of-own-distribution threshold, which is meaningful on both datasets.

Q5 approach — engineer for scale vs. simple baseline. Naive reuse of Q2's per-impression BM25 scoring on the large test sets extrapolated to ~47–50 hours. Given the choice between a scalable-but-uninteresting popularity baseline and actually engineering BM25 to run at scale, we chose the latter (§4) — it produces a submission that reflects the real method, not a fallback.

3. Experimental Observations

3.1 Candidate generation (Q2/Q3): recall@200 over the full catalog

Dataset	BM25	XLM-R (raw)	mpnet (fine-tuned)
EB-NeRD (Danish, 20,738 articles)	2.01%	0.95%	1.69%
MIND (English, 65,238 articles)	1.43%	0.37%	2.27%

Two findings. First, fine-tuning roughly *doubles* recall over the raw checkpoint on both datasets — un-fine-tuned mean-pooled transformer embeddings are a known-weak retrieval baseline (this is the entire reason Sentence-BERT-style fine-tuning exists), and our own numbers confirm it directly. Second, the BM25-vs-embeddings winner **flips by language**: BM25 wins Danish, mpnet wins English — plausibly because mpnet's multilingual capability is distilled *from* an English-only teacher, so its English quality likely exceeds its Danish quality.

3.2 Official ranking task (Q4): re-ranking each impression's own shown candidates

Metric	EB-NeRD BM25	EB-NeRD Emb. (XLM-R)	MIND BM25	MIND Emb. (XLM-R)
AUC	0.497	0.526	0.545	0.557
MRR	0.319	0.334	0.282	0.276
nDCG@10	0.432	0.449	0.318	0.315
Coverage	0.936	0.478	0.982	0.620

The winner **reverses from §3.1**. Candidate generation is "find rare needles in a 20K–65K article haystack" — exact term matching wins there. Re-ranking is "distinguish among ~15–30 already-similar candidates" — a subtler task, where even the *raw*, un-fine-tuned embedding beats BM25 outright on EB-NeRD and is competitive on MIND. This is a genuine architectural finding: **a two-stage system (BM25 to generate candidates, embeddings to re-rank) plays to both methods' actual strengths**, rather than picking one as strictly superior.

Coverage tells the opposite story on both datasets: BM25 touches 58–96% more of the catalog than embeddings. Exact-term queries vary a lot across users and naturally spread recommendations; mean-pooled embeddings repeatedly cluster around the same central/popular articles — a real filter-bubble risk the accuracy metrics alone never surface.

Cold vs. warm (25th-percentile split): AUC drops for cold users in 3 of 4 runs, as expected (less personalization signal) — the exception, EB-NeRD BM25, is already ~random overall (0.497), so there's little further to degrade.

3.3 Real Codabench leaderboard (Q5, large test sets — the actual held-out ground truth)

Metric	MIND BM25	MIND Embeddings (mpnet)
AUC	0.5675	0.6218

EB-NeRD's leaderboard scores are not available to report. `predictions.zip` was submitted on 2026-08-24; as of writing (six days later) Codabench still shows its status as **"Submitted"** rather than "Finished" or "Failed", well past the "up to a few hours" the competition's own guidelines promise. We verified this is not an error on our side before concluding it's a platform-side delay: re-checked zip integrity (`zipfile.testzip()`, no corruption), re-confirmed the required internal filename (`predictions.txt`), and re-ran the 2,000-row random-sample format validation against the original submission -- all clean. With no error log or failure state to act on, there is nothing further to fix locally; this is recorded here as the honest state of that submission rather than a fabricated or extrapolated number.

MIND's real result is nonetheless the strongest evidence in the whole report: on its real 2.37M-impression held-out test set, embeddings **beat** BM25 by a full 0.054 AUC — the same direction as §3.2's offline finding, now confirmed against genuine ground truth. It also validates generalization: MIND BM25's real AUC (0.5675) came in *above* our offline small-split estimate (0.545), meaning the pipeline behaves consistently at 69x scale, not as an artifact of the development split.

Search...	Q	Status			
ID # ▾	File name	Date	Status	Score	Actions
905985	mind_prediction_embeddings.zip	2026-08-28 18:55	Finished	0.6218	✓ ↻
899715	mind_prediction.zip	2026-08-24 17:39	Finished	0.5675	📄 ↻ 🗑

Fig. 1 — MIND Codabench leaderboard. mind_prediction_embeddings.zip scored 0.6218, mind_prediction.zip (BM25) scored 0.5675.

4. Anti-Gaming (Q9): metrics with vs. without a serving-time-unavailable feature

Q2/Q3 never use a leaky feature to begin with, so there was nothing to compare against without deliberately constructing one. We blended BM25's score with $\log_{1p}(\text{article popularity})$, computed either **safe** (click counts from TRAIN only, which precedes val/test in our split) or **LEAKY** (click counts including val+test's own clicks — the period being evaluated):

Metric	EB-NeRD BM25	EB-NeRD +safe pop.	EB-NeRD +LEAKY pop.	MIND BM25	MIND +safe pop.	MIND +LEAKY pop.
AUC	0.497	0.469	0.543	0.545	0.553	0.564

The leaky variant beats the safe variant on every metric, both datasets, with non-overlapping 95% CIs across 71,631/34,519 impressions — real, statistically clear metric inflation, not noise. The size of the inflation tracks the underlying data: EB-NeRD's popularity distribution shifts hugely from train-only to train+val+test (median clicked-article count 17→74), so the gap is large (+0.073 AUC); MIND's shift is much smaller (median 2→3), so the gap is smaller (+0.011) but still clearly present. Severity of a leakage bug depends on the data, not just its existence. (Full numbers: results/anti_gaming_comparison.md.) Leak-freedom of the actual pipeline is enforced and tested in tests/test_no_leakage.py, including an integration check against the real generated EB-NeRD/MIND splits, not just synthetic examples.

5. Where It Breaks at 10× Scale

We hit this wall directly, not hypothetically. Q2's BM25 evaluation ran at ~70–80 impressions/s on the 34K–72K-row small test splits. The large test sets are 69–190x bigger (2.37M / 13.5M impressions); naive extrapolation put full scoring at **47–50 hours** — infeasible.

Root cause: score_candidates() originally called _score_all(), which scans every document sharing a term with the query, then discards all but the ~15–30 requested candidates. A query built from 10 titles routinely contains common words with postings lists in the thousands — wasted work at any catalog size, but only *fatal* at 10x+. Fix: restructure to score the given candidates directly (§2), cutting cost from $O(\text{corpus documents matching any query term})$ to $O(\text{candidates} \times \text{query terms})$ — benchmarked ~1000x faster, reducing the full run to ~15 min. The second lever, per-user query caching, only worked because we *verified* (not assumed) that EB-NeRD's test history ends at an exact one-second boundary before the test period starts — a correctness check that had to happen before trusting the speed optimization.

Other things that would need to change past this point: (1) the FAISS **flat** index is exact but linear in catalog size — a further 10x (approaching MIND's full 161K-article catalog or EB-NeRD's full news archive) would need an approximate index (IVF/HNSW) to stay fast at query time; (2) mean-pooling a user's whole recent history into one vector already loses nuance for users with diverse interests — at 10x more history per user this dilution gets worse, not better; (3) EB-NeRD's/MIND's history being a **fixed pre-period snapshot** means personalization freshness is capped regardless of scale — a production system would need incrementally updated user profiles, which neither dataset's schema supports as-is.

6. AI Usage

Full prompt-level log with verification steps at ai_usage_log.md.